

MCT 211

Automatic Control

Lecture 1: Introduction to Control Systems

Mechatronics Dep.
Faculty of Engineering
Ain Shams University

By Dr. Diaa Emad

Course Grading Scheme

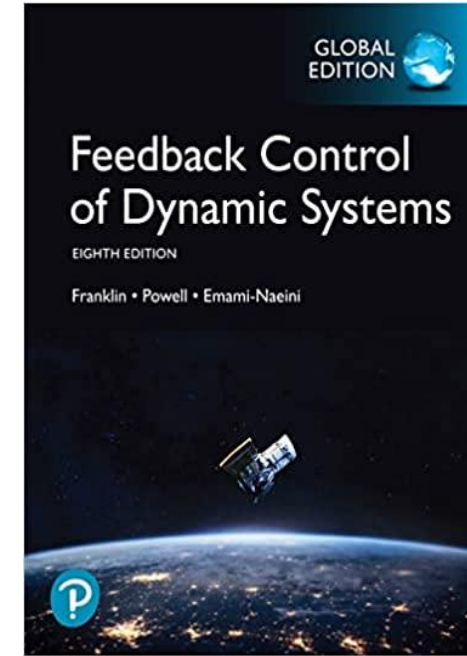
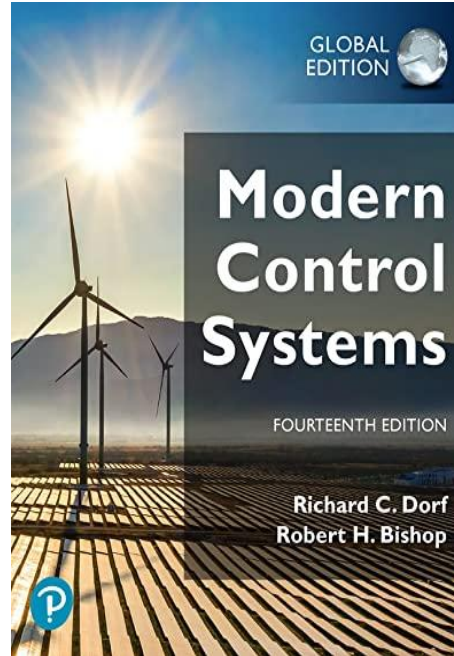
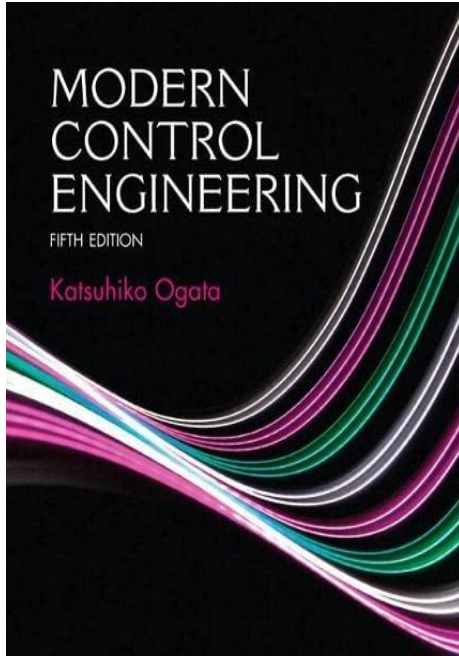
Quiz	5 %
Lap Reports	10 %
Assignments	5 %
Mini-Projects	20 %
Midterm Exam	20 %
Final Exam	40 %
Total	100 %

Course Plan

Lecture	3 Hours/week
Tutorial	1 Hours/week
Lab	1 Hours/week

Lab Software: MATLAB/Simulink

Course References

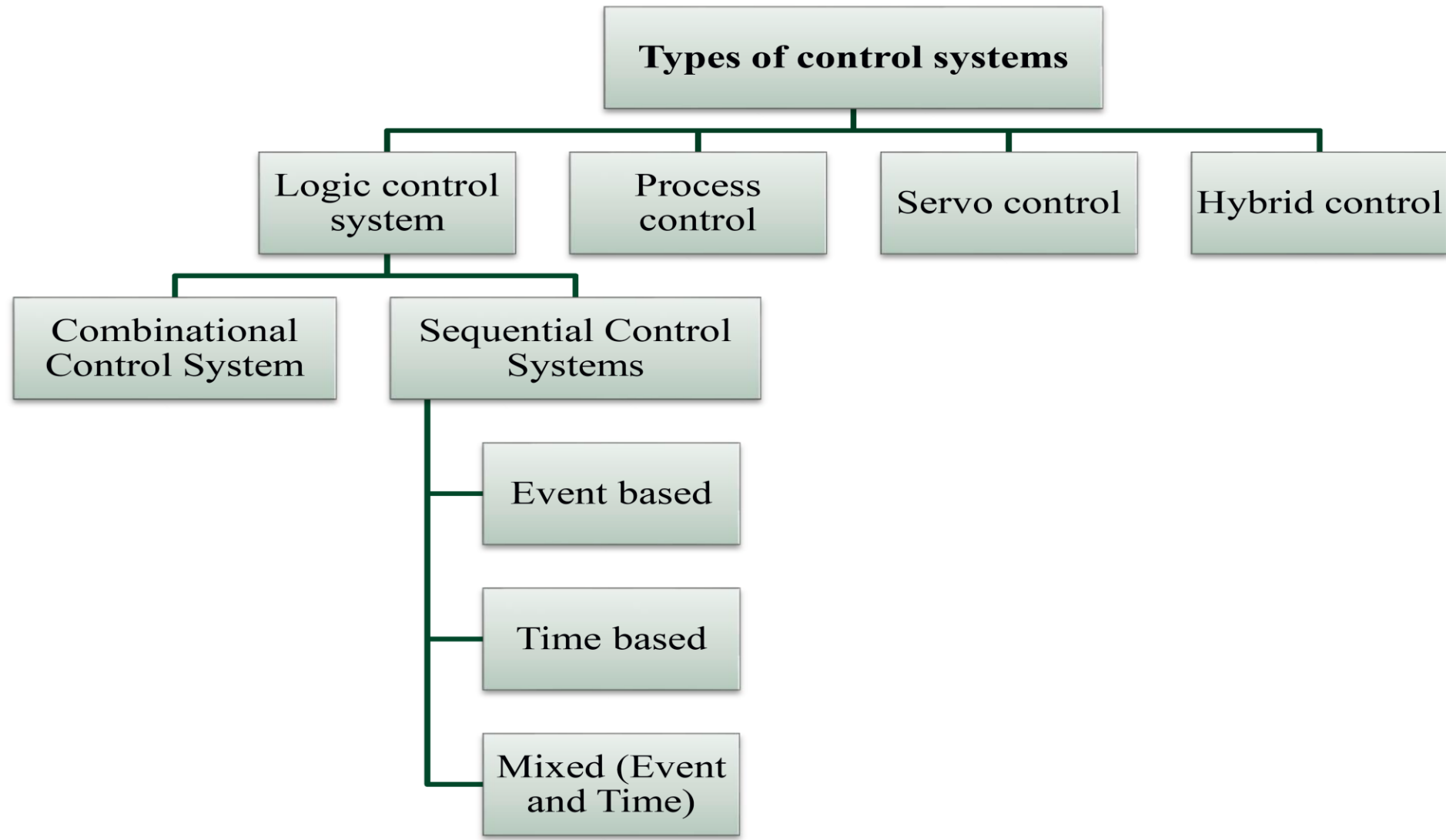


1. Katsuhiko Ogata, “***Modern Control Engineering***”, 5th Edition, Prentice Hall, 2010
2. Richard C. Dorf, Robert H. Bishop, “***Modern Control Systems***”, 14th Edition, Pearson, 2022
3. Gene F. Franklin, J. David Powell, Abbas Emami-Naeini, “***Feedback Control of Dynamic Systems***”, 8th Edition, Pearson, 2018

Course Contents

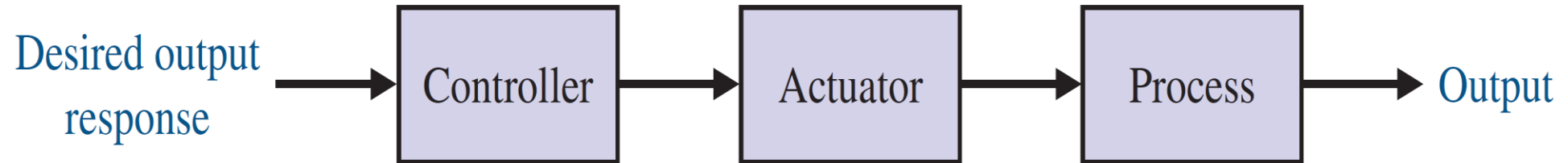
1. Introduction to Control Systems
2. Mathematical Modeling of Control Systems
3. Time Response Analysis
4. Frequency Response Analysis
5. PID Control Design
6. Stability and Steady State Error
7. Root Locus Analysis
8. Lead/Lag Compensators Design

Types of Control Systems



Introduction to Control Systems

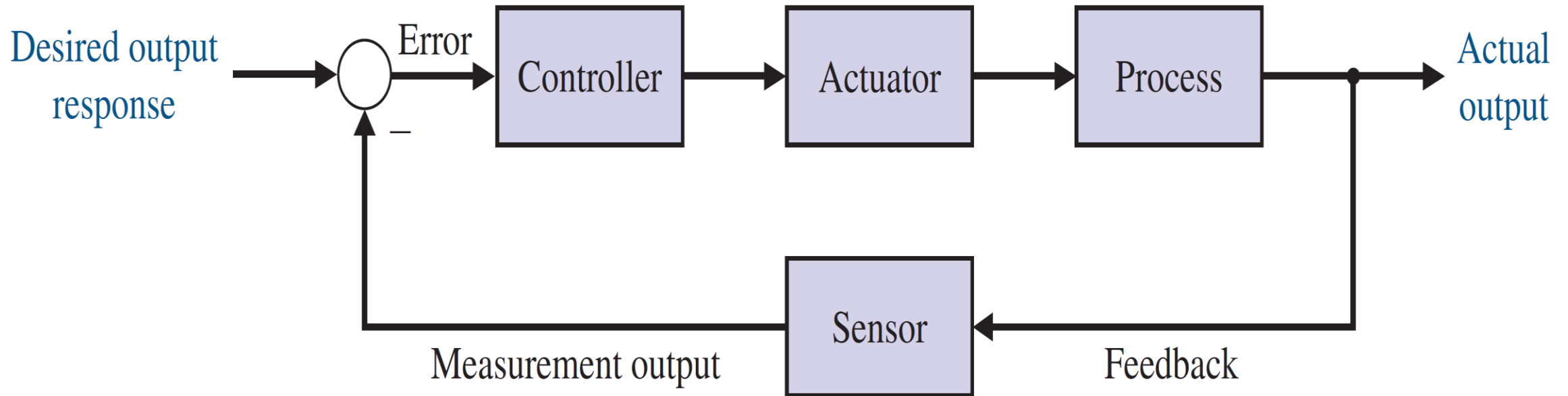
Open Loop Control System



No Guarantee to reach reference

Introduction to Control Systems

Closed Loop Control System

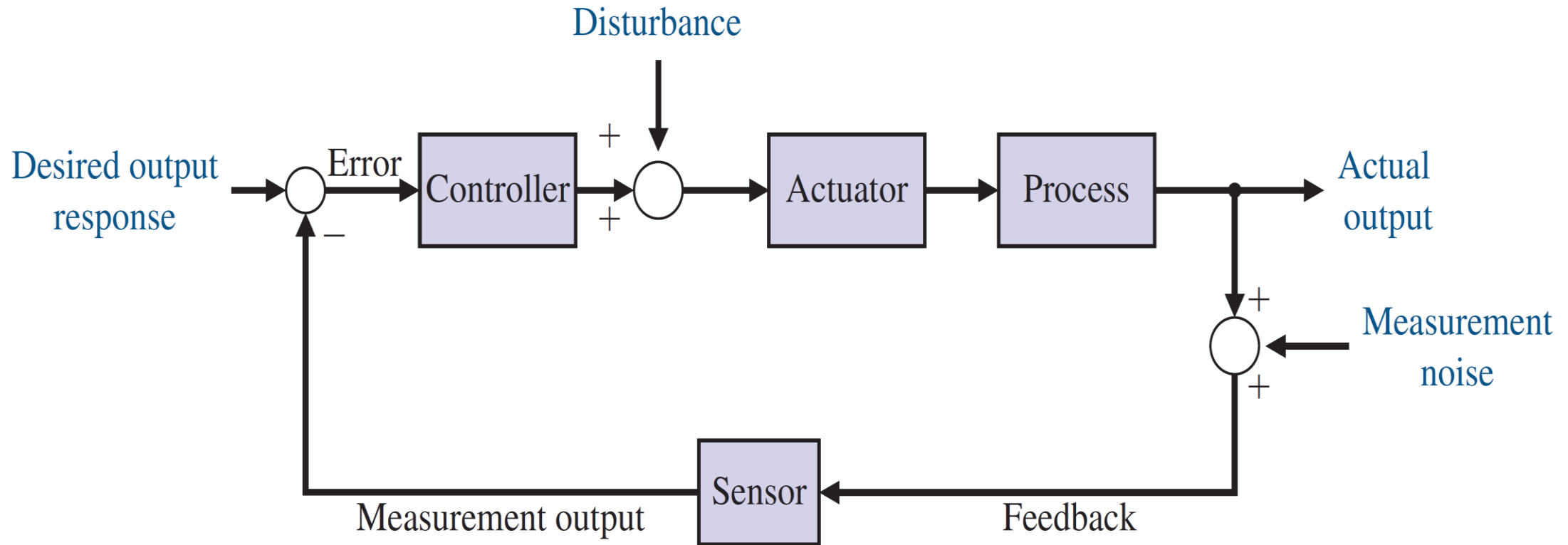


Reference Tracking

Single Input Single Output (SISO)

Introduction to Control Systems

Closed Loop Control System



Disturbance Rejection

Single Input Single Output (SISO)

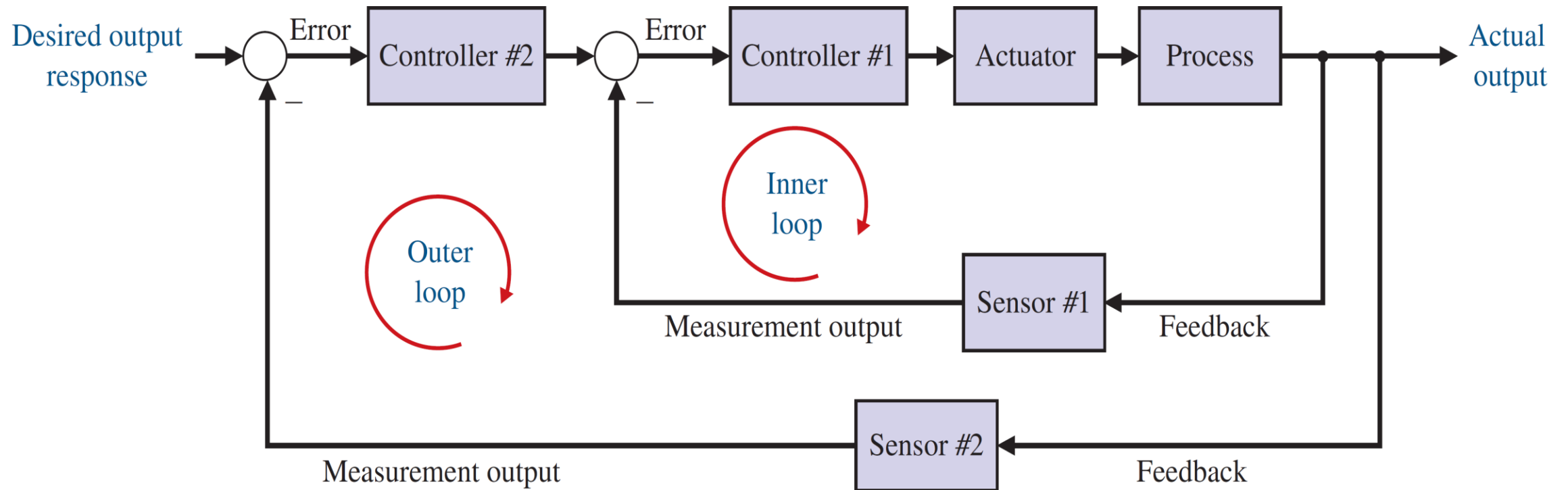
Introduction to Control Systems

Control Systems Terminology

- **The Plant:** It is the body, process or machine of which a particular variable or quantity is to be controlled.
- **Controlled Variable:** The variable to be controlled.
- **Disturbance:** Undesired external excitation which affects the value of the controlled variable.
- **Control Action:** The plant is affected by a signal “ μ ” coming from the controller side.
- **Measuring Device:** It contains an element which should be physically affected by the value of the controlled variable. This element is known by “Sensors”. When it is required to transform the energy of the sensor signal from one to another, an additional element may be used termed by “Transducer”.
- **Controller:** It is the device responsible for establishing certain function relation between the error signal “E” and the control signal “u” .
- **Final Control Element:** It is a powerful device, which is responsible for adjusting the values of the control signal coming from the controller.

Introduction to Control Systems

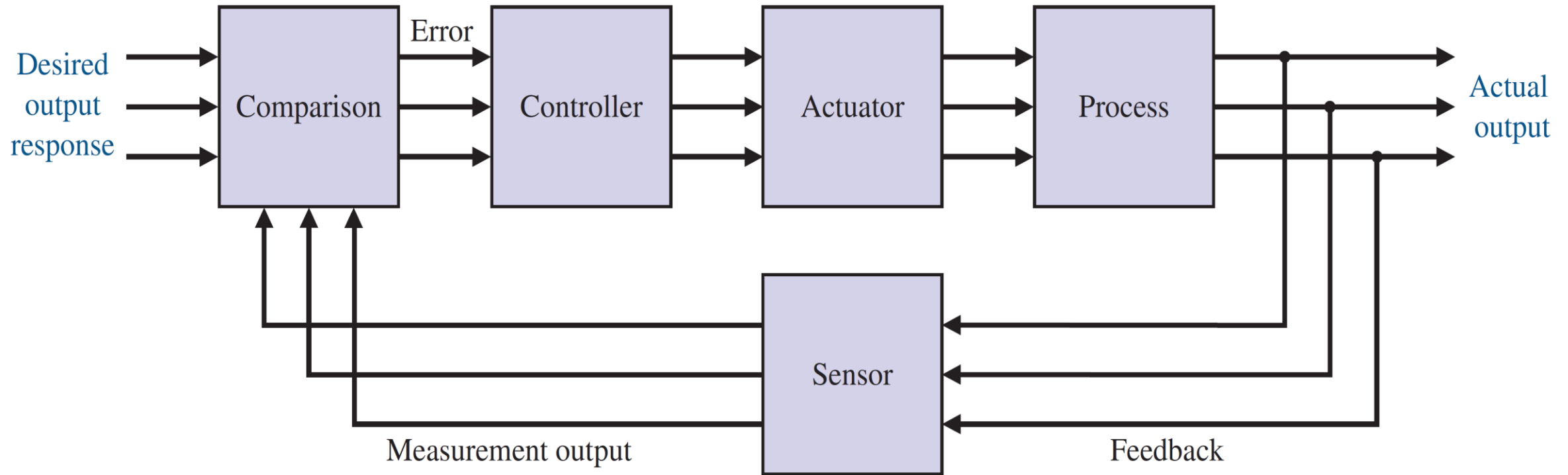
Closed Loop Control System



Multi-loop feedback system with
an inner loop and an outer loop

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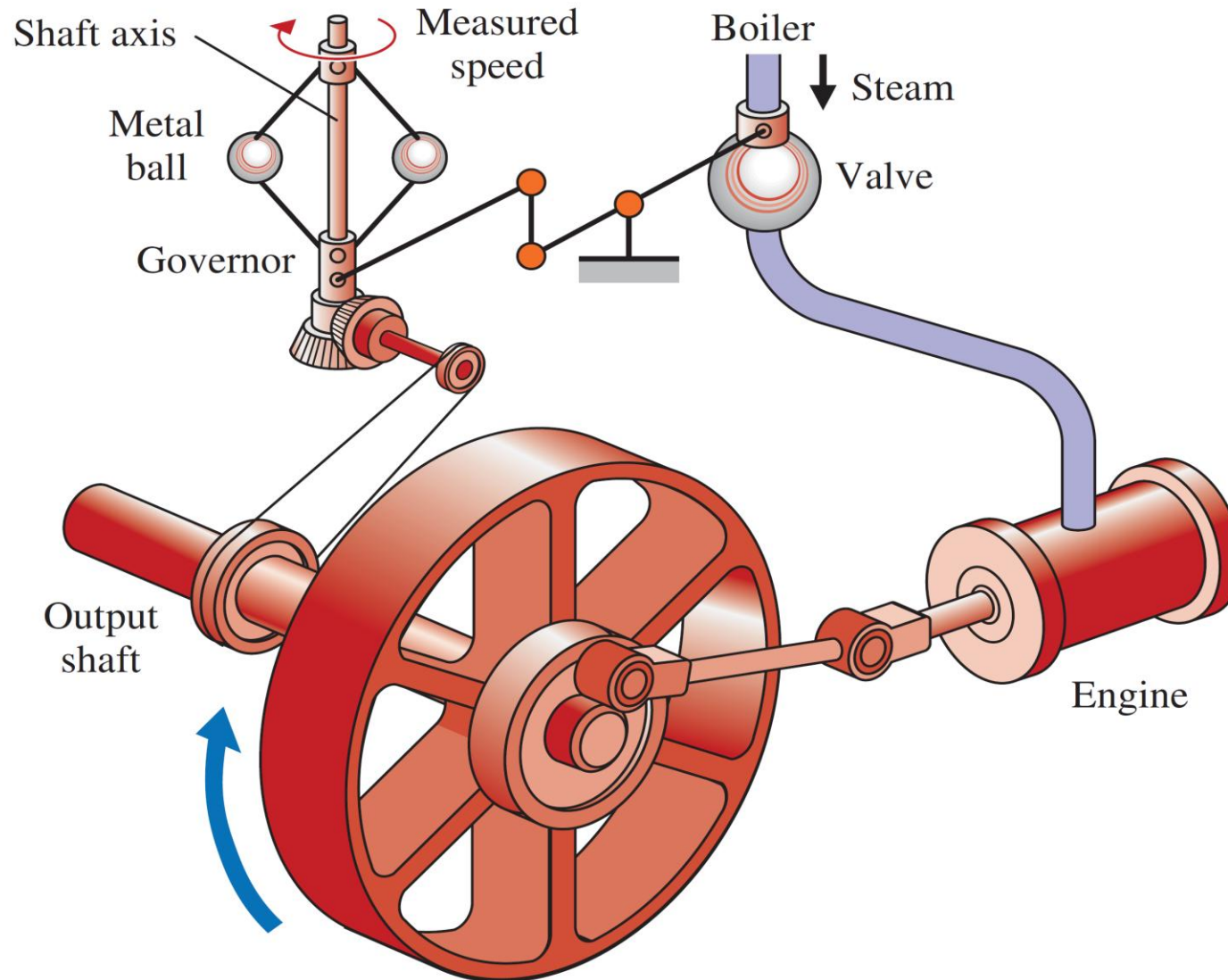
Closed Loop Control System



Multivariable control system
Multi Input Multi Output (MIMO)

Introduction to Control Systems

Speed Control of Steam Engine

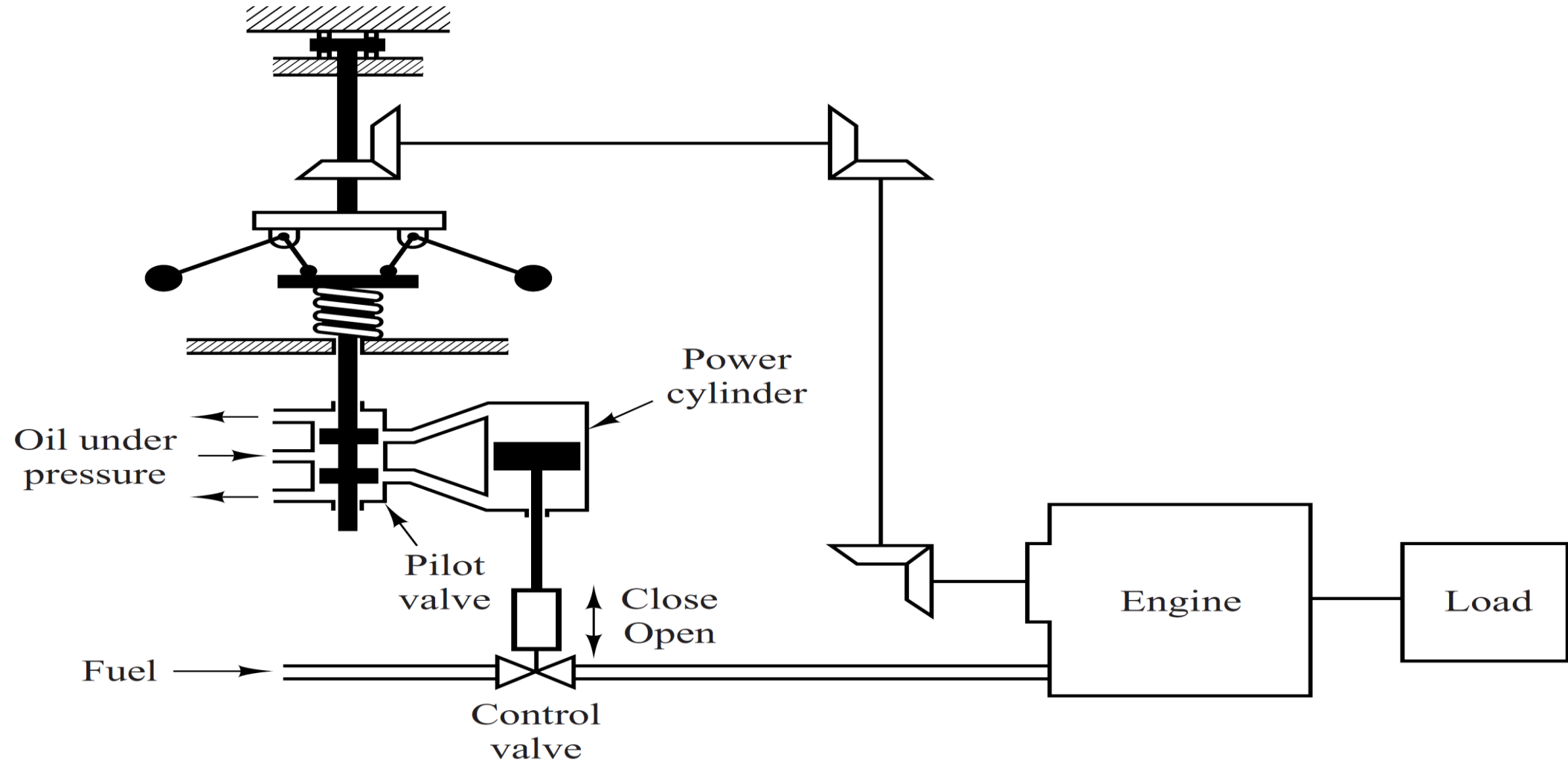


James Watt's
Flyball Governor

The first automatic
feedback controller

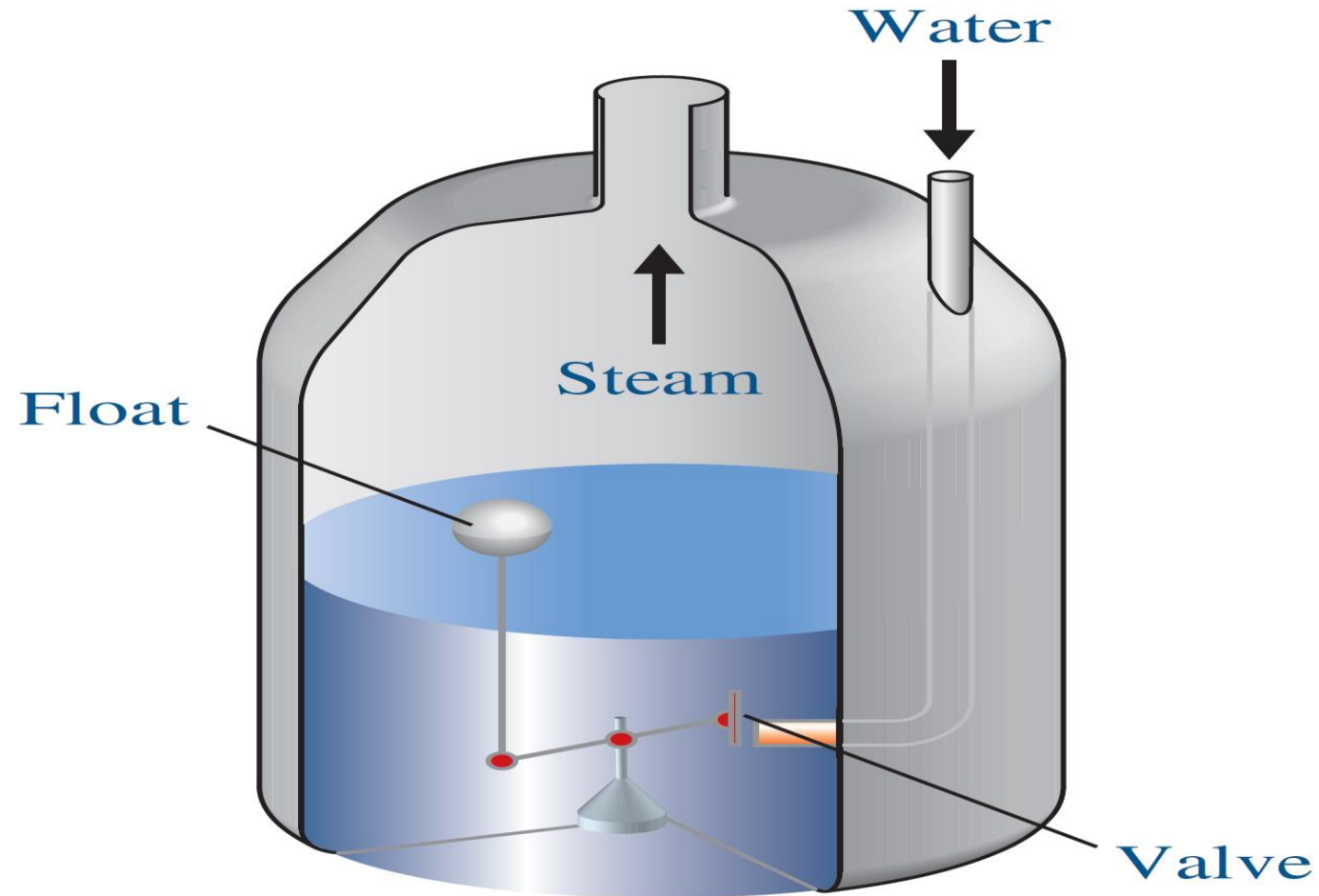
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Speed Control of Fuel Engine



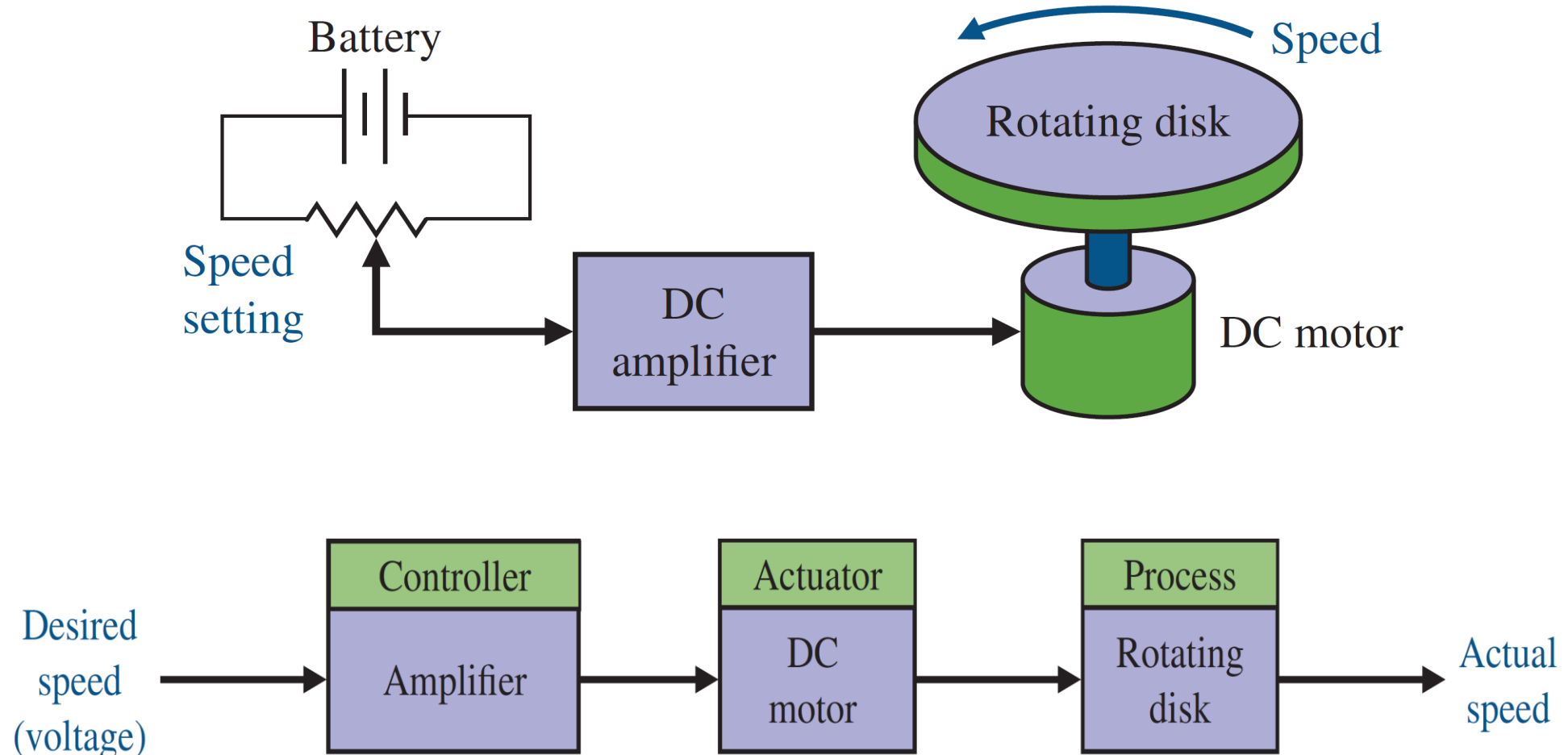
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Water-level float Regulator



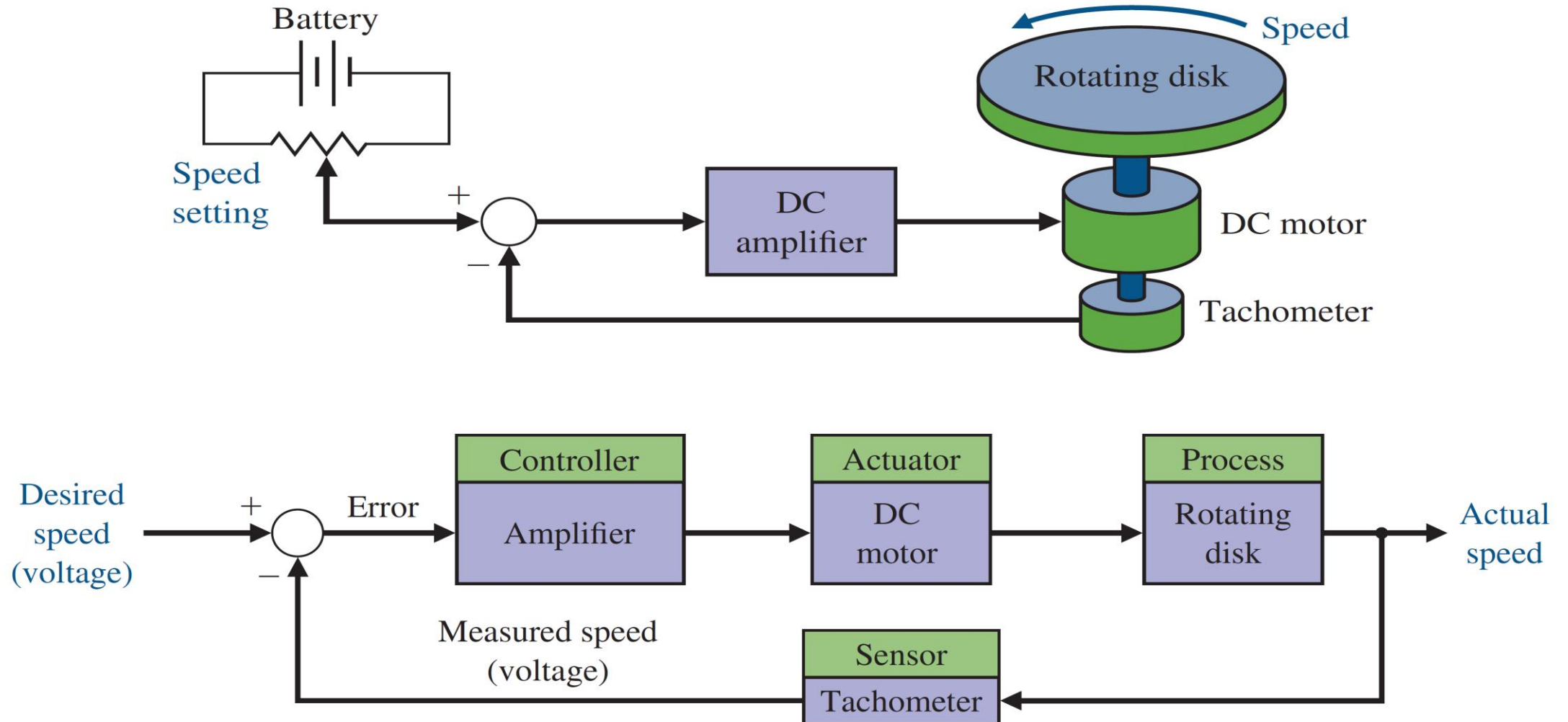
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Speed Control of Rotating Disk



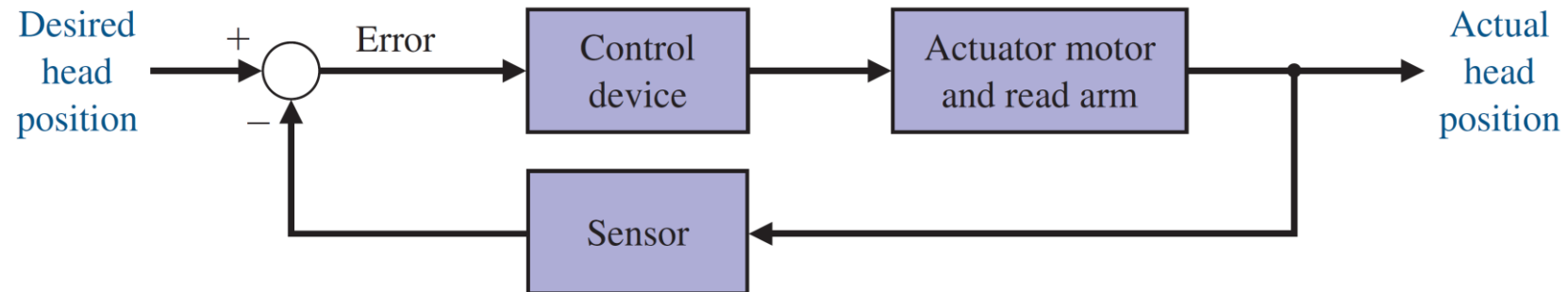
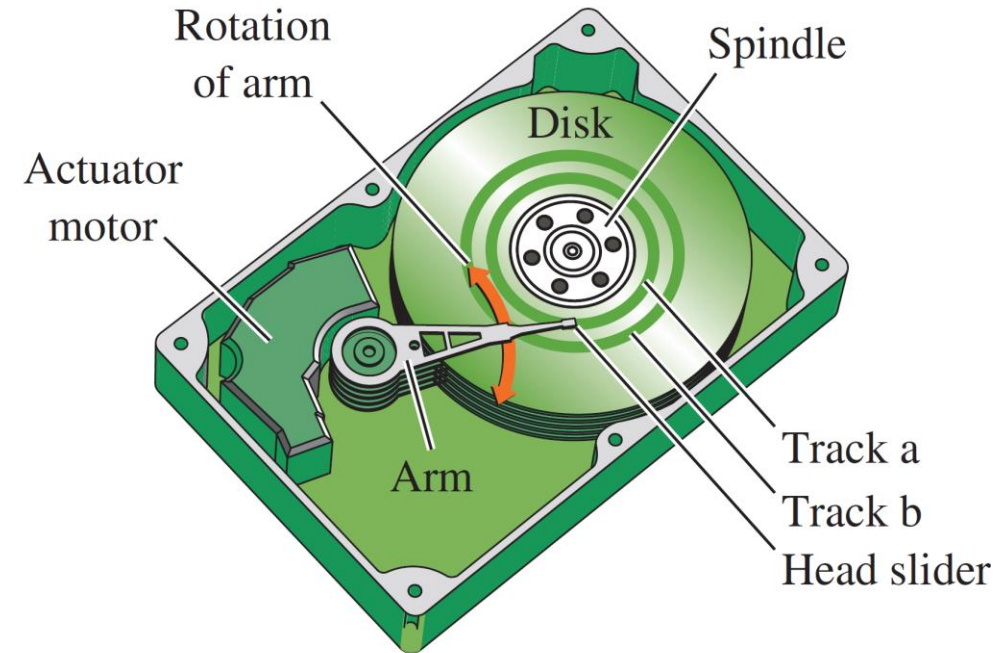
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Speed Control of Rotating Disk



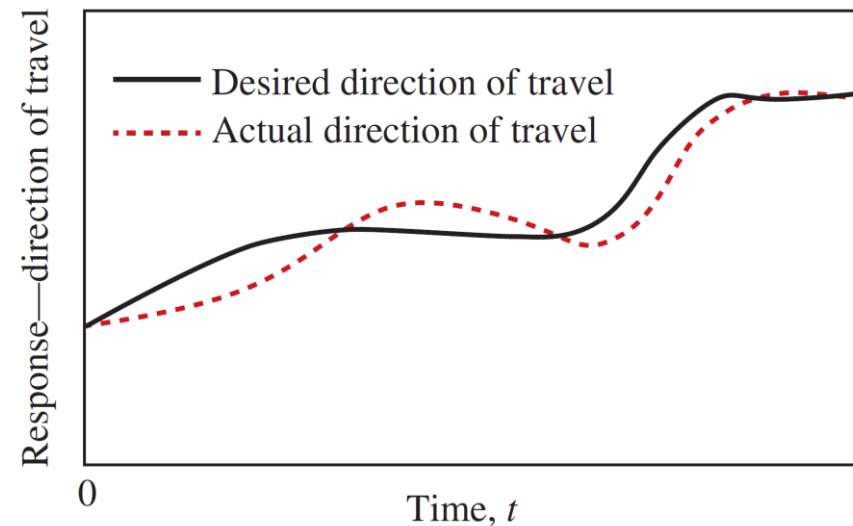
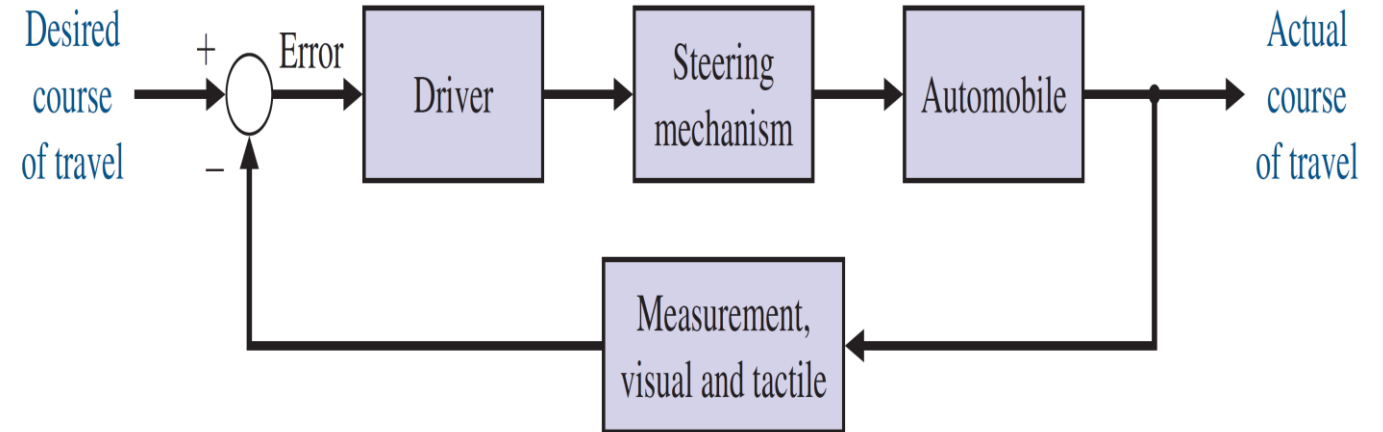
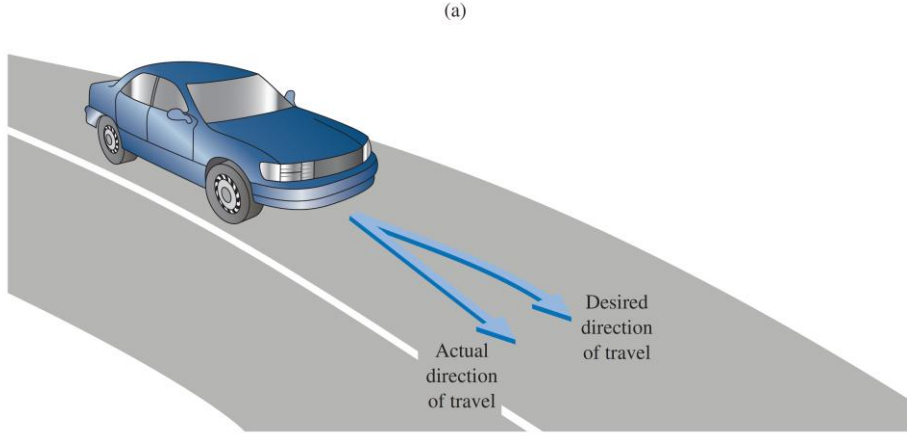
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Position Control of Hard Disk Drive (HDD)



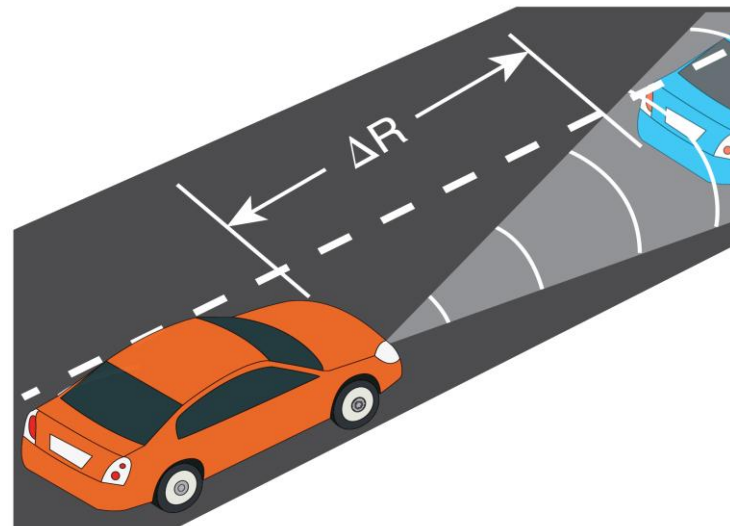
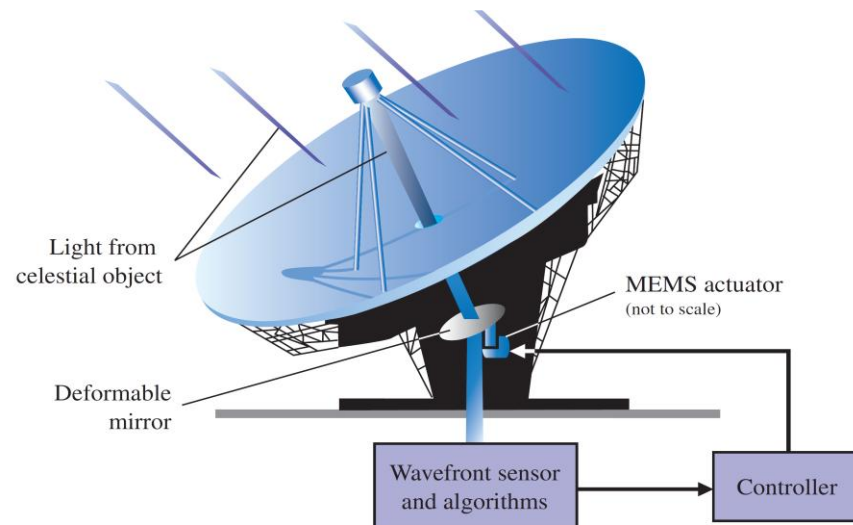
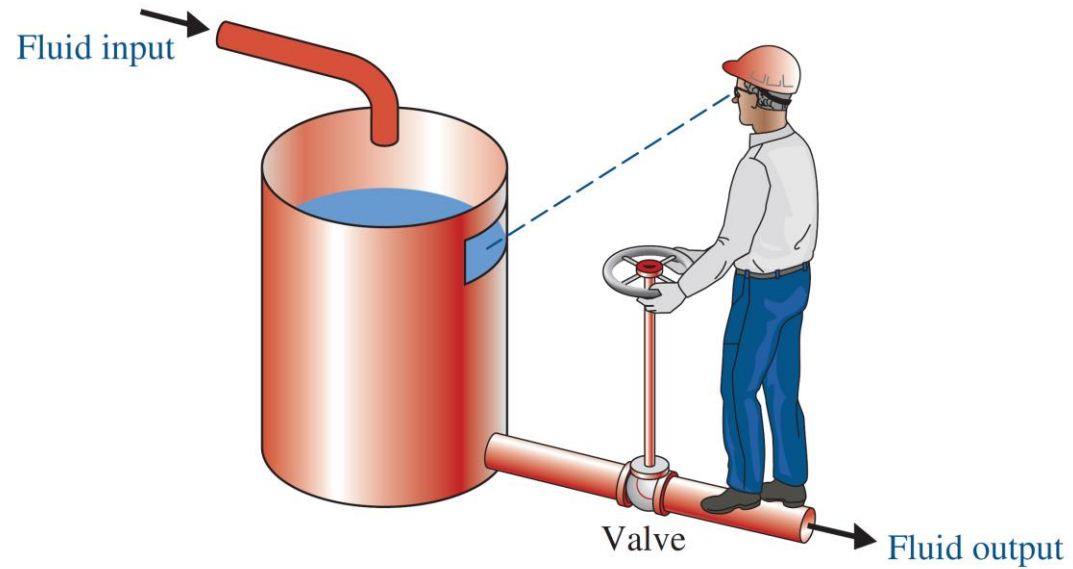
Introduction to Control Systems

Other Examples



Introduction to Control Systems

Other Examples



Thank You

Any Questions ??